THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** metre	SI unit of length is metre (m).
2	**(d)** thread	A thread follows the curve.
3	**(b)** changes with time	Change of position.
4	**(c)** 100 centimetres	1 m = 100 cm.
5	**(d)** directly above the mark	Avoids parallax error.
6	**(d)** the seat of the train	Relative to the seat, no change.
7	**(b)** 1000 metres	1 km = 1000 m.
8	**(d)** parallax error	Reading from an angle gives parallax.
9	**(b)** the person is moving	Rest and motion are relative.
10	**(c)** 10 millimetres	1 cm = 10 mm.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Everyone gets the same value (1); Results can be compared and shared (1).	1+1
12	Lay a thread along the path (1); Straighten the thread and measure it (1).	1+1
13	They differ from person to person (1); Results will not agree (1).	1+1
14	1 m = 100 cm (1); 2.5 x 100 = 250 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Place the zero of the ruler at one end (1); Keep your eye above the other end (1); Read the mark and write with unit (1).	1+1+1
16	Zero may be missing (1); It gives wrong length (1); Use a clear full mark (1).	1+1+1
17	Place zero mark at the start (1); Read from above (1); Use a full mark if the end is broken (1).	1+1+1
	OR: 4 km = 4000 m (1); Add 250 m (1); Total = 4250 m (1).	1+1+1
18	Different lengths for different people (1); Not accurate (1); Cannot be compared (1).	1+1+1
	OR: Body parts used (1); Cubit, span (1); Standard units made later (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Linear: a car, a falling stone (1); Circular: stone whirled on a string, hands of a clock (1); Rotational: top, fan (1); Periodic: pendulum, swing (1); Random: butterfly (1).	1+1+1+1+1
	OR: Measure 100 sheets together (1); Total thickness is 1.2 cm (1); Divide by 100 (1); Gives 0.012 cm (1); Convert to 0.12 mm (1).	1+1+1+1+1
20	Place thread on the curve (1); Mark the end (1); Straighten and measure (1); Wind thread once round bottle (1); Measure the thread (1).	1+1+1+1+1
	OR: Cubit: elbow to fingertip (1); Foot: length of the foot (1); Pace: step length (1); Vary from person to person (1); Metre introduced as standard (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up conversions: 1 m = 100 cm (not 10 cm), 1 cm = 10 mm, 1 km = 1000 m.
- Reading the scale from the side, which gives a wrong value; keep the eye straight above the mark.
- Starting the measurement from the 1 cm mark without subtracting; either start at zero or subtract the first reading.
- Stretching or loosening the thread while measuring a curved line.
- Calling a spinning top **circular** motion; spinning about its own axis is rotational.
- Saying all repeating motion is periodic; it must repeat after **equal** intervals of time.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** start measuring from a full mark like 1 cm and subtract it	Use a clear mark and subtract.
2	**(c)** rotational	Rotation about its axis.
3	**(a)** handspan	Handspan varies from person to person.
4	**(b)** at one end of the pencil	Zero aligns with one end.
5	**(b)** circular (orbital)	It revolves along an orbit.
6	**(b)** tip of the middle finger	Cubit was elbow to fingertip.
7	**(c)** units	Units give meaning.
8	**(a)** a butterfly flying	Motion without a fixed pattern.
9	**(b)** to get the same measurement whoever measures	Standard units remove confusion.
10	**(c)** linear (rectilinear)	It moves in a straight line.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Linear: car on a straight road (1); Circular: a stone tied to a string and whirled (1).	1+1
12	1 cm = 10 mm (1); 5 x 10 = 50 mm (1).	1+1
13	1 km = 1000 m (1); 3500 / 1000 = 3.5 km (1).	1+1
14	Motion repeating at regular time intervals (1); Swing or pendulum (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	6.4 x 100 = 640 cm (1); 640 x 10 = 6400 mm (1); Both from conversions (1).	1+1+1
16	Passenger moves w.r.t. the platform (1); At rest w.r.t. the train seat (1); Motion depends on the observer (1).	1+1+1
17	Lay the thread on the line (1); Mark the end (1); Measure the straight thread (1).	1+1+1
	OR: Measure thickness of 100 sheets together (1); Divide by 100 (1); Gives thickness of one sheet (1).	1+1+1
18	We are moving (1); They are fixed to the ground (1); Relative motion (1).	1+1+1
	OR: Metre (1); 1 m = 100 cm = 1000 mm (1); 1 km = 1000 m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Body units vary (1); Standard unit is the same for all (1); SI unit is metre (1); Multiples: km, cm, mm (1); Conversions are easy (1).	1+1+1+1+1
	OR: Pendulum moves to and fro (1); Repeats at equal intervals (1); Time for 20 = 40 s (1); 40 / 20 (1); 2 s per oscillation (1).	1+1+1+1+1
20	Parallax: wrong eye position (1); Zero at one end (1); Eye straight over mark (1); Read and write the value (1); Include unit (1).	1+1+1+1+1
	OR: Spinning top: rotational (1); Car: linear (1); Swing: periodic (1); Bird: random (1); Hand of a clock: circular (1).	1+1+1+1+1

Mistakes to avoid:

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- Starting the measurement from the 1 cm mark without subtracting; either start at zero or subtract the first reading.
- Stretching or loosening the thread while measuring a curved line.
- Calling a spinning top **circular** motion; spinning about its own axis is rotational.
- Saying all repeating motion is periodic; it must repeat after **equal** intervals of time.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** periodic motion	E.g. a pendulum.
2	**(d)** ruler	A scale or ruler measures short lengths.
3	**(b)** circular	It moves along a circle.
4	**(c)** rotational	Fan rotates.
5	**(a)** measuring tape	Flexible tapes measure body lengths.
6	**(c)** periodic (to and fro)	It repeats at equal intervals.
7	**(d)** millimetre	A very small length.
8	**(d)** kilometres	Long distances use km.
9	**(b)** linear	Straight-line path.
10	**(a)** waist of a person	Curved/circular things.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is flexible (1); It can bend around the body (1).	1+1
12	Error from reading at a slant (1); Keep the eye directly above the mark (1).	1+1
13	Position does not change with time (1); It is relative to something else (1).	1+1
14	Spinning top (1); Ceiling fan (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Linear: straight line, a running athlete (1); Circular: along a circle, a stone tied to a string (1); Both can be observed (1).	1+1+1
16	Pendulum: periodic (1); Wheel: rotational (1); Ball: linear (1).	1+1+1
17	Depends on the observer (1); Same object may seem to move or rest (1); Example: a bus passenger (1).	1+1+1
	OR: Periodic: pendulum, swing, clock hand (1); Non-periodic: flying bird (1); Running dog (1).	1+1+1
18	1.5 m (1); 750 m (1); 4.8 cm (1).	1+1+1
	OR: Walking: linear (1); Spinning: rotational (1); Vibrating string: periodic (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) 3600 m (1); (b) 7.25 m (1); (c) 240 cm (1); 240 / 6 (1); 40 cm (1).	1+1+1+1+1
	OR: Keep tape straight (1); Start from zero mark (1); Add repeated lengths (1); Keep eye above reading (1); Record with unit (1).	1+1+1+1+1
20	Passenger in bus (1); At rest relative to seat (1); Moving relative to the road (1); Moon seems to move with us (1); Observer decides (1).	1+1+1+1+1
	OR: Train passenger (1); Platform observer (1); Trees seem to move backward (1); Passenger sees the co-passenger at rest (1); Relative motion (1).	1+1+1+1+1

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